

EXPERIMENT 11

MODES OF DES-ALL TYPES

AIM

To implement the different modes of DES (ECB, CBC, CFB, OFB, and CTR) using a simple C program.

ALGORITHM

1. Read the plaintext.
2. Read a single-character key.
3. Select the DES mode.
4. Perform encryption according to the selected mode.
5. Display the encrypted text.
6. End the program.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char text[100], key;
    int i, choice;

    printf("Enter Plain Text: ");
    fgets(text, sizeof(text), stdin);
    text[strcspn(text, "\n")] = '\0';

    printf("Enter Key (Single Character): ");
    scanf(" %c", &key);

    printf("\n1.ECB\n2.CBC\n3.CFB\n4.OFB\n5.CTR");
    printf("\nEnter Choice: ");
    scanf("%d", &choice);

    printf("\nEncrypted Text: ");

    for(i=0;text[i]!='\0';i++)
    {
        switch(choice)
        {
```

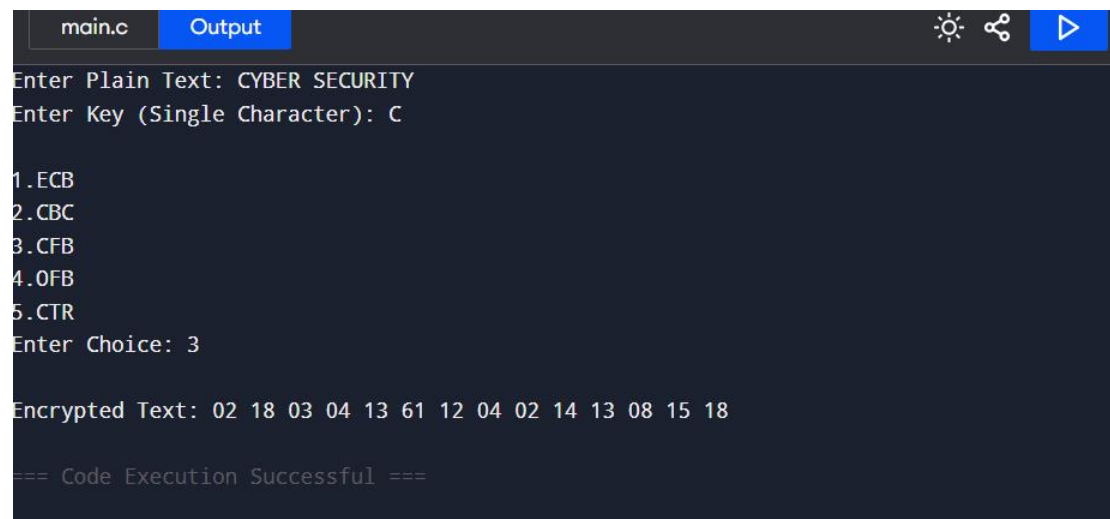
```

        case 1: printf("%02X ", text[i]^key); break;
        case 2: printf("%02X ", (text[i]^key)^1); break;
        case 3: printf("%02X ", (text[i]^key)^2); break;
        case 4: printf("%02X ", (text[i]^key)^3); break;
        case 5: printf("%02X ", (text[i]^key)^i); break;
        default: printf("Invalid Choice");
                return 0;
    }
}

return 0;
}

```

OUTPUT



The screenshot shows a terminal window titled 'main.c' with an 'Output' tab. The program prompts the user to enter plain text and a key. The plain text 'CYBER SECURITY' and key 'C' are entered. A menu of encryption modes is displayed: 1.ECB, 2.CBC, 3.CFB, 4.OFB, and 5.CTR. Choice 3 (CFB) is selected. The resulting encrypted text is shown as a sequence of hexadecimal values: 02 18 03 04 13 61 12 04 02 14 13 08 15 18. The execution concludes with the message '=== Code Execution Successful ==='.

```

main.c  Output
Enter Plain Text: CYBER SECURITY
Enter Key (Single Character): C

1.ECB
2.CBC
3.CFB
4.OFB
5.CTR
Enter Choice: 3

Encrypted Text: 02 18 03 04 13 61 12 04 02 14 13 08 15 18

=== Code Execution Successful ===

```

RESULT

The different DES modes (ECB, CBC, CFB, OFB, and CTR) were successfully simulated and different encrypted outputs were obtained for each mode.